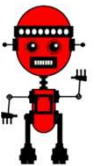
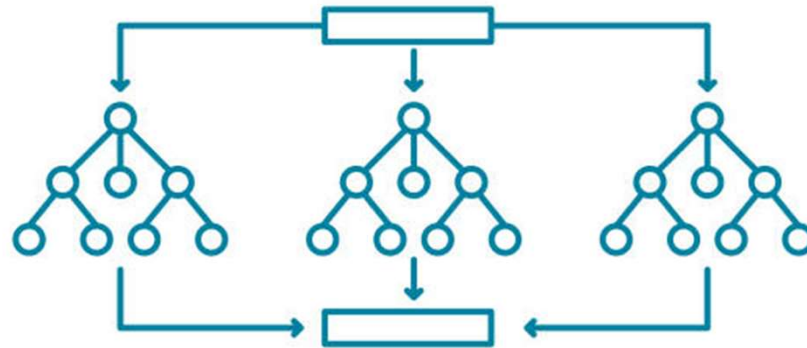


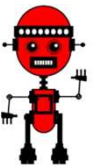
Decision Tree Machine learning Algorithm



CONTENT



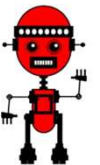
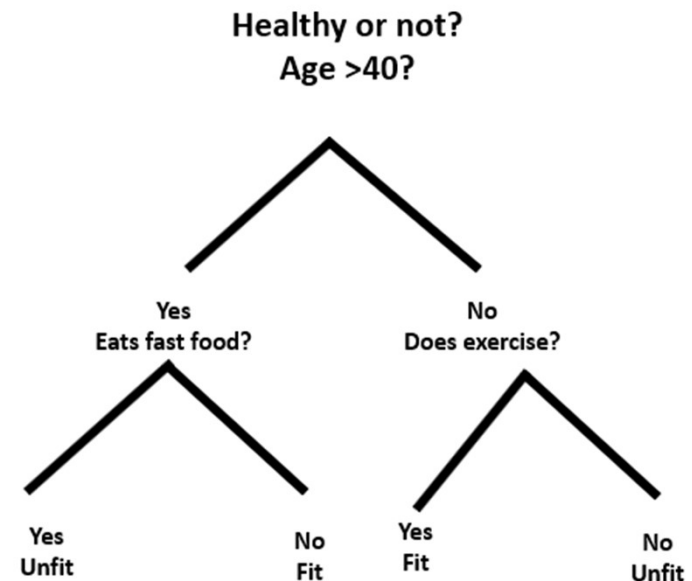
- Concept of Decision Tree
- Terminology related to DT
- Attribute Selection measure
- Pruning
- Merits and Demerits



Concept of Decision Tree

Employes divide and conquer strategy by conducting a greedy search to identify the optimal split points within a tree.

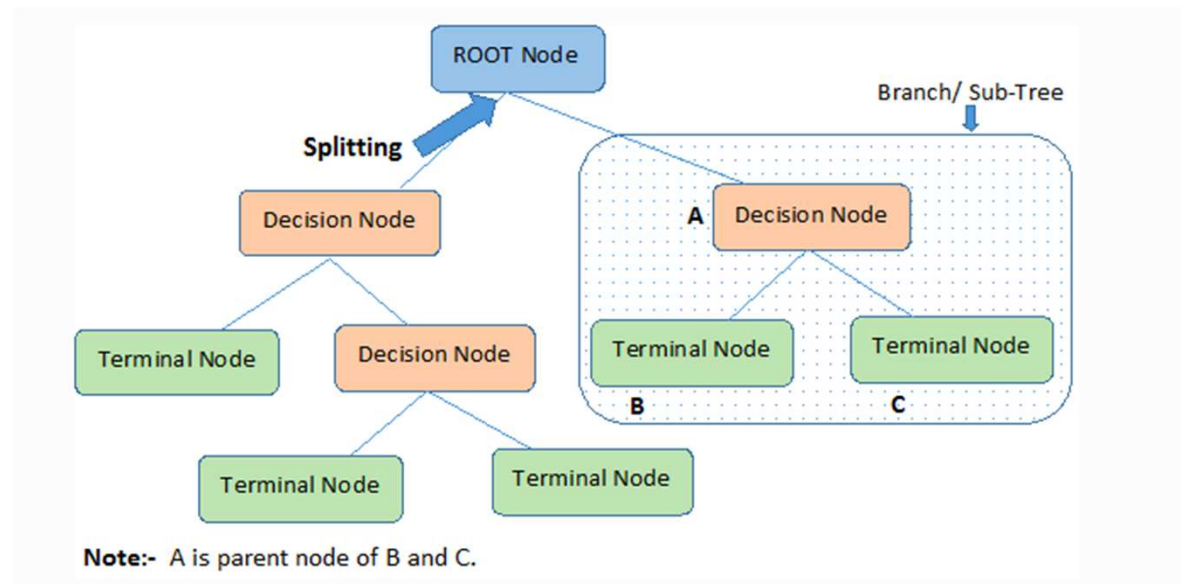
- Non parametric supervised machine learning algorithm
- Used for both regression and classification problems
- Based on hierarchical tree structure



Workflow

- Formation of node for each attribute
- Selection of root node (best attribute)
- Splitting start based on ASM
- Splitting continue till leaf node
- Where finally we get prediction

Important terminology



Attribute Selection Measure

Information gain

$$Gain(S, A) \equiv Entropy(S) - \sum_{v \in D_A} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

- A represents a specific attribute or class label
- $Entropy(S)$ is the entropy of dataset, S
- $|S_v|/|S|$ represents the proportion of the values in S_v to the number of values in dataset, S
- $Entropy(S_v)$ is the entropy of dataset, S_v

Gini Index

$$\begin{aligned} Gini\ Impurity &= 1 - \sum_{i=1}^K p_i^2 \\ &= 1 - Gini\ Index \end{aligned}$$

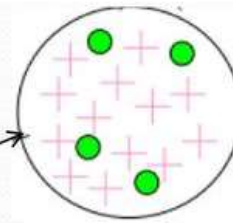
where K is the number of class labels,

p_i is the proportion of i^{th} class label

$$\text{Information Gain} = \text{entropy}(\text{parent}) - [\text{average entropy}(\text{children})]$$

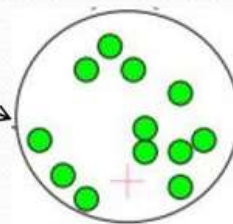
Entire population (30 instances)

$$\text{parent entropy} = -\left(\frac{14}{30} \cdot \log_2 \frac{14}{30}\right) - \left(\frac{16}{30} \cdot \log_2 \frac{16}{30}\right) = 0.996$$



17 instances

$$\text{child entropy} = -\left(\frac{13}{17} \cdot \log_2 \frac{13}{17}\right) - \left(\frac{4}{17} \cdot \log_2 \frac{4}{17}\right) = 0.787$$



13 instances

$$\text{child entropy} = -\left(\frac{1}{13} \cdot \log_2 \frac{1}{13}\right) - \left(\frac{12}{13} \cdot \log_2 \frac{12}{13}\right) = 0.391$$

$$(\text{Weighted}) \text{ Average Entropy of Children} = \left(\frac{17}{30} \cdot 0.787\right) + \left(\frac{13}{30} \cdot 0.391\right) = 0.615$$

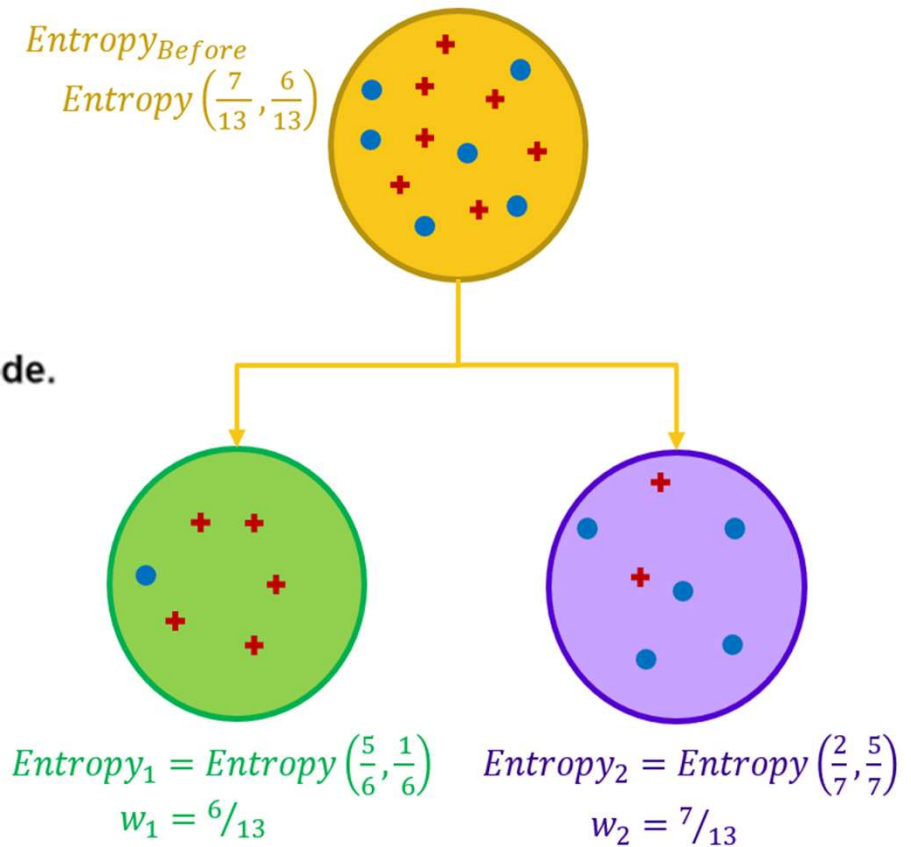
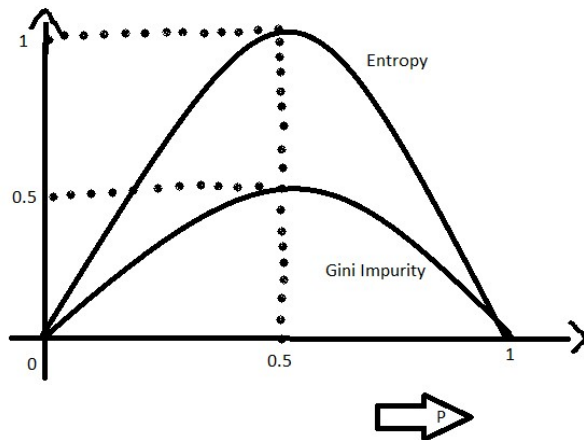
$$\text{Information Gain} = 0.996 - 0.615 = 0.38$$

Parameters which measure impurity in dataset

$$Gini = 1 - \sum_{i=1}^n p^2(c_i)$$

$$Entropy = \sum_{i=1}^n -p(c_i) \log_2(p(c_i))$$

where $p(c_i)$ is the probability/percentage of class c_i in a node.

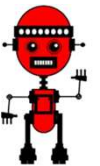
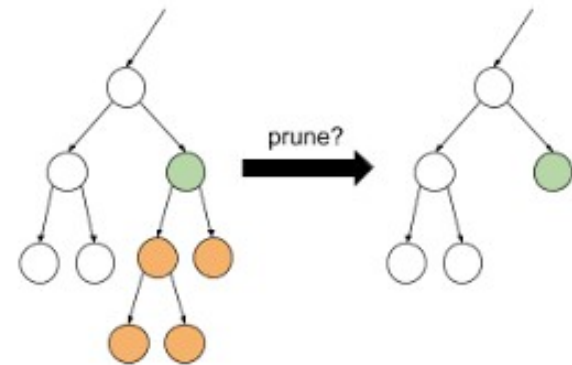


How to avoid risk of overfitting?

Pruning reduces the complexity of the final classifier, and hence improves predictive accuracy by the reduction of overfitting.

Pruning should reduce the size of a learning tree without reducing predictive accuracy as measured by a cross-validation set.

- Pre-Pruning
- Post-Pruning



Frequently Applied

- To handling non-linear data sets effectively.
- engineering, civil planning, law, and business.

Merits and Demerits

- Simple to understand
- Solves Decision related problem
- Less requirement of Data cleaning
- Less appropriate for estimation tasks where the goal is to predict the value of a continuous attribute.
- Prone to errors in classification problems with many classes and a relatively small number of training examples.
- Computationally expensive to train.

